

Cauchy-Schwarz Masterclass

Notes

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1. Starting with Cauchy

1.1. Proofs

Proposition 1.1.1 (Cauchy-Schwarz): For $a_i, b_i \in \mathbb{R}$ we have

$$a_1 b_1 + a_2 b_2 + \cdots + a_n b_n \leq \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2} \sqrt{b_1^2 + b_2^2 + \cdots + b_n^2}.$$

Proof of Cauchy-Schwarz (via Induction):

Let $H(n)$ stand for the hypothesis that Cauchy's inequality is valid for n .

The base case $H(1)$ is trivially true as the $a_1 b_1 = \sqrt{a_1^2} \sqrt{b_1^2}$. The second base case $H(2)$ follows from observing that,

$$\begin{aligned} 0 &\leq (a_1 b_2 - b_1 a_2)^2 \\ &\implies 2a_1 a_2 b_1 b_2 \leq a_1^2 b_2^2 + b_1^2 a_2^2 \\ &\implies a_1^2 b_1^2 + 2a_1 a_2 b_1 b_2 + a_2^2 b_2^2 \leq a_1^2 b_2^2 + a_1^2 b_1^2 + a_2^2 b_2^2 + b_1^2 a_2^2 \\ &\implies a_1 b_1 + a_2 b_2 \leq \sqrt{a_1^2 + a_2^2} \sqrt{b_1^2 + b_2^2}. \end{aligned}$$

Now, assuming $H(n)$ holds for some $n \geq 1$ we wish to show that $H(n+1)$ also holds. Consider, using the inductive hypothesis,

$$a_1 b_1 + a_2 b_2 + \cdots + a_n b_n + a_{n+1} b_{n+1} \leq \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2} \sqrt{b_1^2 + b_2^2 + \cdots + b_n^2} + a_{n+1} b_{n+1}.$$

Then, using $H(2)$,

$$\leq \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2 + a_{n+1}^2} \sqrt{b_1^2 + b_2^2 + \cdots + b_n^2 + b_{n+1}^2}.$$

Thus, $H(n+1)$ also holds. By induction, Cauchy's inequality holds for all $n \in \mathbb{N}$. □

Lemma 1.1.1:

$$\sum_{k=1}^{\infty} a_k^2 < \infty \text{ and } \sum_{k=1}^{\infty} b_k^2 < \infty \text{ implies that } \sum_{k=1}^{\infty} |a_k b_k| < \infty.$$

Proof (without Cauchy-Schwarz): We want to show that $a_k b_k$ is small whenever a_k^2 and b_k^2 are small. The following observation gives us what we want,

$$0 \leq (x - y)^2 \implies xy \leq \frac{1}{2}(x^2 + y^2).$$

We apply this to $x = |a_k|$ and $y = |b_k|$ and add the inequalities up,

$$\sum_{k=1}^{\infty} |a_k b_k| \leq \frac{1}{2} \sum_{k=1}^{\infty} a_k^2 + \frac{1}{2} \sum_{k=1}^{\infty} b_k^2 < \infty. \quad (1)$$

□

Proof of Cauchy-Schwarz (via eqn. (1)): Assume that neither $\{a_k\}$ nor $\{b_k\}$ are made of zeroes. Then define the sequences,

$$\hat{a}_k = \frac{a_k}{\left(\sum_{k=1}^{\infty} a_k^2\right)^{\frac{1}{2}}}, \quad \hat{b}_k = \frac{b_k}{\left(\sum_{k=1}^{\infty} b_k^2\right)^{\frac{1}{2}}}.$$

Now, apply eqn. (1) to $\{\hat{a}_k\}$ and $\{\hat{b}_k\}$.

$$\begin{aligned} \sum_{k=1}^{\infty} \hat{a}_k \hat{b}_k &\leq \frac{1}{2} \sum_{k=1}^{\infty} \hat{a}_k^2 + \frac{1}{2} \sum_{k=1}^{\infty} \hat{b}_k^2 = 1 \\ \Rightarrow \sum_{k=1}^{\infty} \frac{a_k b_k}{\left(\sum_{k=1}^{\infty} a_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^{\infty} b_k^2\right)^{\frac{1}{2}}} &\leq 1 \\ \Rightarrow \sum_{k=1}^{\infty} a_k b_k &\leq \left(\sum_{k=1}^{\infty} a_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^{\infty} b_k^2\right)^{\frac{1}{2}}. \end{aligned}$$

Remark: Normalization is a systematic way of getting from an *additive inequality* to a *multiplicative inequality*. □

Proposition 1.1.2: Equality holds in the Cauchy-Schwarz inequality iff the sequences $\{a_k\}$ and $\{b_k\}$ are scalar multiple of one another.

Proof: We focus on the nontrivial case where neither of the sequences is identically zero and where both $\sum_{k=1}^{\infty} a_k^2, \sum_{k=1}^{\infty} b_k^2$ are finite.

The backward direction is easy to prove by a routine computation. We focus on the forward direction. The equality

$$\sum_{k=1}^{\infty} a_k b_k = \left(\sum_{k=1}^{\infty} a_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^{\infty} b_k^2\right)^{\frac{1}{2}} \quad (2)$$

implies the equality (with $\hat{a}_k, \hat{b}_k$ defined as above)

$$\sum_{k=1}^{\infty} \hat{a}_k \hat{b}_k = \frac{1}{2} \sum_{k=1}^{\infty} \hat{a}_k^2 + \frac{1}{2} \sum_{k=1}^{\infty} \hat{b}_k^2 = 1. \quad (3)$$

By the two-term bound, $xy \leq \frac{1}{2}(x^2 + y^2)$ we also know that

$$\hat{a}_k \hat{b}_k \leq \frac{1}{2} \hat{a}_k^2 + \frac{1}{2} \hat{b}_k^2 \quad \text{for all } k = 1, 2, \dots,$$

If any of these inequalities were strict, then we wouldn't get the equality in eqn. (3). Thus, the equality in eqn. (2) holds for a nonzero series only when we have $\hat{a}_k = \hat{b}_k$ for all $k = 1, 2, \dots$. By the definition of these normalized values, we have that

$$a_k = \lambda b_k \quad \text{for all } k = 1, 2, \dots,$$

with λ given by the ratio

$$\lambda = \left(\sum_{j=1}^{\infty} a_j^2\right)^{\frac{1}{2}} / \left(\sum_{j=1}^{\infty} b_j^2\right)^{\frac{1}{2}}.$$

□

1.2. Notation and Generalizations

The Cauchy-Schwarz inequality can be quite compactly represented in the context of an *inner product space*. We introduce the requisite material here.

Definition 1.2.1: Suppose V is a real vector space. Then a function on $V \times V$ defined by the mapping $(\mathbf{a}, \mathbf{b}) \mapsto \langle \mathbf{a}, \mathbf{b} \rangle$ is an *inner product* and we say that $(V, \langle \cdot, \cdot \rangle)$ is a *real inner product space* provided that the pair $(V, \langle \cdot, \cdot \rangle)$ has the following properties

- a. $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$ for all $\mathbf{v}, \mathbf{w} \in V$
- b. $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ for all nonzero $\mathbf{v} \in V$
- c. $\langle \alpha \mathbf{v} + \mathbf{u}, \mathbf{w} \rangle = \alpha \langle \mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$ for all $\alpha \in \mathbb{R}$ and $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$

On $\mathbb{R}^n$, the following inner product is popular.

Definition 1.2.2 (Euclidean Inner Product): For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$,

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{j=1}^n a_j b_j$$

This lets us rewrite the Cauchy-Schwarz inequality succinctly.

Proposition 1.2.1: For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ we have

$$\langle \mathbf{a}, \mathbf{b} \rangle \leq \langle \mathbf{a}, \mathbf{a} \rangle^{\frac{1}{2}} \langle \mathbf{b}, \mathbf{b} \rangle^{\frac{1}{2}}.$$

Of course, there are other inner product spaces too!

Example: On $\mathbb{R}^n$ the following weighted sum defines an inner product,

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{j=1}^n a_j b_j w_j.$$

Example: Consider the vector space $C[a, b]$ of real-valued continuous functions on the bounded interval $[a, b]$. Then, for any continuous $w : [a, b] \rightarrow \mathbb{R}$ such that $w(x) > 0$ for all $x \in [a, b]$, we can define an inner product on $C[a, b]$ by setting

$$\langle f, g \rangle = \int_a^b f(x)g(x)w(x) dx.$$

This naturally leads us to ask whether Cauchy-Schwarz is true for all inner product spaces.

Proposition 1.2.2: Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Then, for any $\mathbf{v}, \mathbf{w} \in V$ we have

$$\langle \mathbf{v}, \mathbf{w} \rangle \leq \langle \mathbf{v}, \mathbf{v} \rangle^{\frac{1}{2}} \langle \mathbf{w}, \mathbf{w} \rangle^{\frac{1}{2}}.$$

For nonzero $\mathbf{v}, \mathbf{w} \in V$ we have

$$\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{v} \rangle^{\frac{1}{2}} \langle \mathbf{w}, \mathbf{w} \rangle^{\frac{1}{2}} \quad \text{if and only if} \quad \mathbf{v} = \lambda \mathbf{w}$$

for a nonzero constant λ .

Proof: We try to use a variant of the additive method developed above. Consider,

$$0 \leq \langle \mathbf{v} - \mathbf{w}, \mathbf{v} - \mathbf{w} \rangle$$

$$\implies \langle \mathbf{v}, \mathbf{w} \rangle \leq \frac{1}{2}(\langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle).$$

We normalize to convert this to a multiplicative bound. Since the inequality holds trivially for $\mathbf{v} = \mathbf{0}$ or $\mathbf{w} = \mathbf{0}$, we assume that $\mathbf{v}, \mathbf{w}$ are nonzero. Define,

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\langle \mathbf{v}, \mathbf{v} \rangle^{\frac{1}{2}}} \quad \hat{\mathbf{w}} = \frac{\mathbf{w}}{\langle \mathbf{w}, \mathbf{w} \rangle^{\frac{1}{2}}}.$$

We then have,

$$\langle \hat{\mathbf{v}}, \hat{\mathbf{w}} \rangle \leq \frac{1}{2}(\langle \hat{\mathbf{v}}, \hat{\mathbf{v}} \rangle + \langle \hat{\mathbf{w}}, \hat{\mathbf{w}} \rangle) = 1$$

$$\implies \left\langle \frac{\mathbf{v}}{\langle \mathbf{v}, \mathbf{v} \rangle^{\frac{1}{2}}}, \frac{\mathbf{w}}{\langle \mathbf{w}, \mathbf{w} \rangle^{\frac{1}{2}}} \right\rangle \leq 1$$

$$\implies \langle \mathbf{v}, \mathbf{w} \rangle \leq \langle \mathbf{v}, \mathbf{v} \rangle^{\frac{1}{2}} \langle \mathbf{w}, \mathbf{w} \rangle^{\frac{1}{2}}.$$

Now, we deal with the necessary condition for equality. If $\mathbf{v}, \mathbf{w}$ are nonzero then the normalized vectors $\hat{\mathbf{v}}, \hat{\mathbf{w}}$ are well defined. Furthermore, equality in the Cauchy-Schwarz inequality then gives us $\langle \hat{\mathbf{v}}, \hat{\mathbf{w}} \rangle = 1$. So,

$$\langle \hat{\mathbf{v}}, \hat{\mathbf{w}} \rangle = \frac{1}{2}(\langle \hat{\mathbf{v}}, \hat{\mathbf{v}} \rangle + \langle \hat{\mathbf{w}}, \hat{\mathbf{w}} \rangle)$$

$$\implies \langle \hat{\mathbf{v}} - \hat{\mathbf{w}}, \hat{\mathbf{v}} - \hat{\mathbf{w}} \rangle = 0$$

$$\implies \hat{\mathbf{v}} = \hat{\mathbf{w}}$$

Thus,

$$\mathbf{v} = \lambda \mathbf{w} \quad \text{for } \lambda = \frac{\langle \mathbf{v}, \mathbf{v} \rangle^{\frac{1}{2}}}{\langle \mathbf{w}, \mathbf{w} \rangle^{\frac{1}{2}}}.$$

□

1.3. Symmetry and Amplification

This material is from [Terence Tao's blog](#).

We consider the general setting of a complex inner product space, V . In this context, the Cauchy-Schwarz inequality is given by

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \langle \mathbf{v}, \mathbf{v} \rangle^{\frac{1}{2}} \langle \mathbf{w}, \mathbf{w} \rangle^{\frac{1}{2}}.$$

To prove this, we start off with the additive bound,

$$0 \leq \langle \mathbf{v} - \mathbf{w}, \mathbf{v} - \mathbf{w} \rangle$$

$$\implies \Re \langle \mathbf{v}, \mathbf{w} \rangle \leq \frac{1}{2}(\langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle).$$

This is a weaker bound than Cauchy-Schwarz as $\Re \langle \mathbf{v}, \mathbf{w} \rangle \leq \langle \mathbf{v}, \mathbf{w} \rangle \leq \langle \mathbf{v}, \mathbf{v} \rangle^{\frac{1}{2}} \langle \mathbf{w}, \mathbf{w} \rangle^{\frac{1}{2}} \leq \frac{1}{2}(\langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle)$. The last inequality follows from the AM-GM inequality (see next chapter for more details).

We can amplify this additive bound by observing some symmetry imbalances. Particularly, the phase rotation $\mathbf{v} \mapsto e^{i\theta} \mathbf{v}$ preserves the right-hand side but not the left-hand side,

$$\Re e^{i\theta} \langle \mathbf{v}, \mathbf{w} \rangle \leq \frac{1}{2}(\langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle).$$

We can choose any real θ we want. To make the left-hand side as large as possible, we choose θ to cancel the phase of $\langle \mathbf{v}, \mathbf{w} \rangle$. This gets us,

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \frac{1}{2}(\langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle).$$

Now, to strengthen the right-hand side we exploit a different symmetry, *homogenisation symmetry*. Particularly, consider the map $(\mathbf{v}, \mathbf{w}) \mapsto (\lambda \mathbf{v}, \frac{1}{\lambda} \mathbf{w})$ for a scalar $\lambda > 0$. This gives us,

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \frac{\lambda^2}{2} \langle \mathbf{v}, \mathbf{v} \rangle + \frac{1}{2\lambda^2} \langle \mathbf{w}, \mathbf{w} \rangle.$$

The choice of $\lambda = \sqrt{\|\mathbf{w}\|/\|\mathbf{v}\|}$ minimizes the right-hand side. This gives us,

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \langle \mathbf{v}, \mathbf{v} \rangle^{\frac{1}{2}} \langle \mathbf{w}, \mathbf{w} \rangle^{\frac{1}{2}}.$$

1.4. Yet Another Proof™

This material is from [Timothy Gower's blog](#). In this section, we will see a more motivated development of a common proof for the Cauchy-Schwarz inequality (it is pretty much the same proof as the one above).

Recall what we mean by the Cauchy-Schwarz result: For $a_k, b_k \in \mathbb{R}$, we have

$$\sum_{k=1}^{\infty} a_k b_k \leq \left(\sum_{k=1}^{\infty} a_k^2 \right)^{\frac{1}{2}} \left(\sum_{k=1}^{\infty} b_k^2 \right)^{\frac{1}{2}}$$

with equality iff the sequences $\{a_i\}$ and $\{b_i\}$ are proportional.

The central idea for our proof will be trying to find a natural way to express the fact that two sequences are proportional. One approach would be to say that there exists a $\lambda \in \mathbb{R}$ such that $a_k = \lambda b_k$ for every k . However, why bother introducing an unknown variable λ unless we absolutely have to? We could simply require all a_k/b_k to be equal. Though, we may be worried about some b_k being zero. We can resolve this by simply saying that two sequences are proportional if $a_k b_j = a_j b_k$ for all j, k .

We want lots of (in fact, for all j, k we want $a_k b_j - a_j b_k = 0$) terms to be zero. This can be expressed by requiring the sum of all their squares to be zero. So, sequences $\{a_k\}$ and $\{b_k\}$ are proportional iff

$$\sum_{k,j} (a_k b_j - a_j b_k)^2 = 0.$$

Also note that the expression on the left is trivially at least zero. By expanding out the left-hand side, we readily obtain both the Cauchy-Schwarz inequality and the necessary condition for equality,

$$\begin{aligned} \sum_{k,j} (a_k b_j - a_j b_k)^2 &= \sum_{k,j} (a_k^2 b_j^2 - 2a_k a_j b_k b_j + a_j^2 b_k^2) \\ &= 2 \sum_{k,j} a_k^2 b_j^2 - 2 \sum_{k,j} a_k b_k a_j b_j \\ &= 2 \sum_k a_k^2 \sum_j b_j^2 - 2 \left(\sum_k a_k b_k \right)^2. \end{aligned}$$

Now, we try to extend this idea to real inner product spaces. We want to show that

$$\langle \mathbf{v}, \mathbf{w} \rangle \leq \|\mathbf{v}\| \|\mathbf{w}\|$$

with equality iff $\mathbf{v}$ and $\mathbf{w}$ are proportional with a positive constant. Again, we motivate our proof by thinking in terms of expressing proportionality. A first attempt is to say that $\mathbf{v}, \mathbf{w}$ are proportional with a positive constant iff $\mathbf{v}/\|\mathbf{v}\| = \mathbf{w}/\|\mathbf{w}\|$ (note how this doesn't work for proportionality in general, for example we could have $\mathbf{v} =$

$-w$). As we did before, we can equivalently express this condition as requiring $\|w\|v - \|v\|w = 0$. So that we may express this using inner products, we consider the squared version of this: $(\|w\|v - \|v\|w)^2 = 0$. Note that the left-hand side is in fact always greater than or equal to zero. Then, expanding the left hand side immediately gives us the Cauchy-Schwarz result,

$$(\|w\|v - \|v\|w)^2 = 2\|v\|^2\|w\|^2 - 2\|w\|\|v\|\langle v, w \rangle.$$

For a complex inner product space, $(\|w\|v - \|v\|w)^2$ expands as

$$(\|w\|v - \|v\|w)^2 = 2\|v\|^2\|w\|^2 - \|w\|\|v\|(\langle v, w \rangle + \langle w, v \rangle)$$

Let x be a complex number with modulus $|x| = 1$ and the property that $\langle w, xv \rangle$ is real and non-negative. Consequently, $\langle w, xv \rangle = |\langle w, v \rangle|$. We readily get that $|\langle w, v \rangle| = \langle w, xv \rangle \leq \|v\|\|w\|$ with equality iff $x\|w\|v - \|v\|w = 0$.

2. The AM-GM Inequality